EXP NO: 10 EXPERIMENT TO UNDERSTAND K-MEANS CLUSTERING ALGORITHM FOR GIVEN DATA SET

AIM:

Understand the K-means clustering algorithm for the dataset given.

PYTHON CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

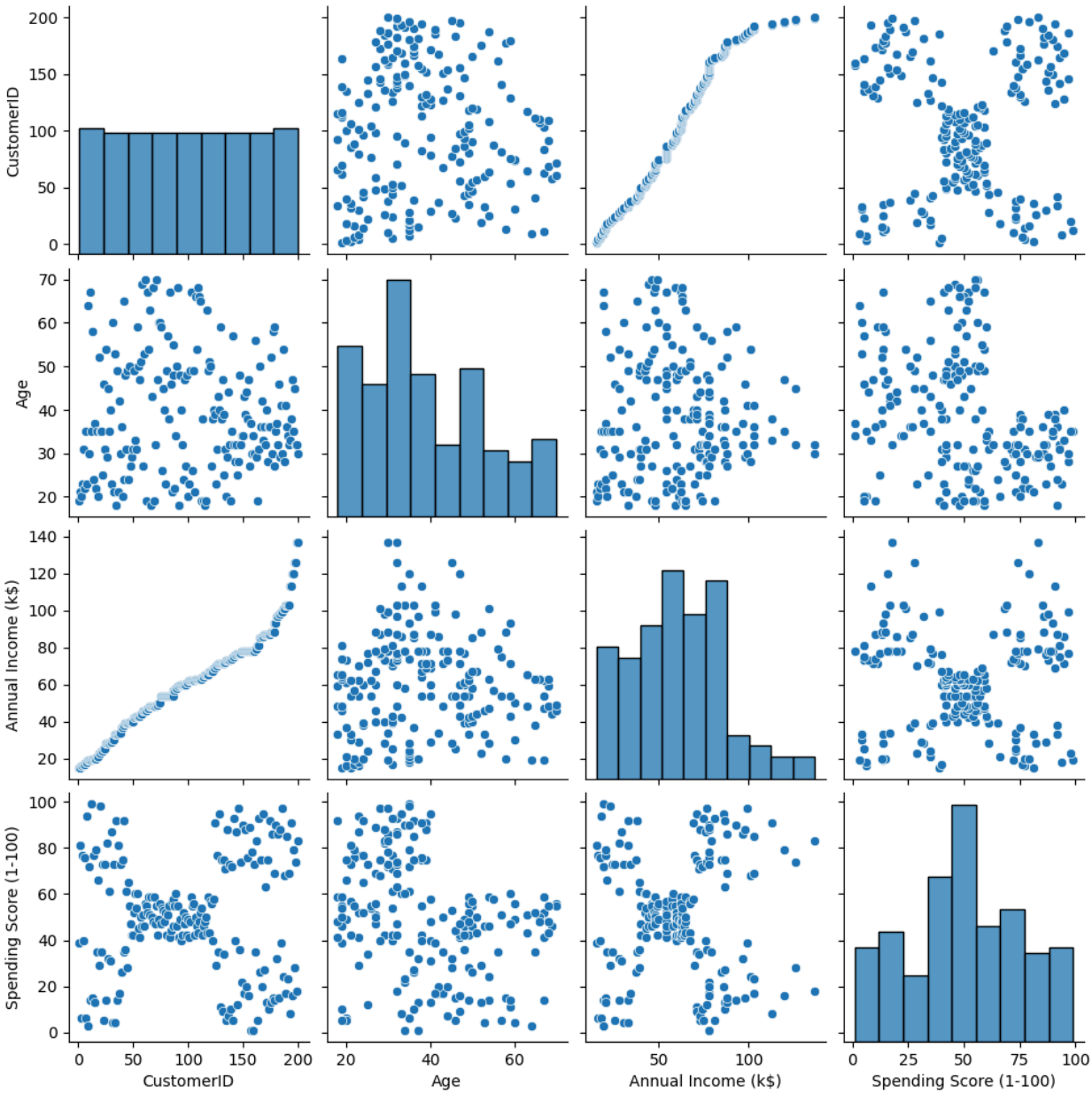
df = pd.read_csv('Mall_Customers.csv')
df.info()
df.head()
sns.pairplot(df)
features = df.iloc[:,[3,4]].values
from sklearn.cluster import KMeans
model = KMeans(n_clusters=5)
model.fit(features)
Final = df.iloc[:,[3,4]]
Final['label'] = model.predict(features)
Final.head()
sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
   .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
   .add_legend()

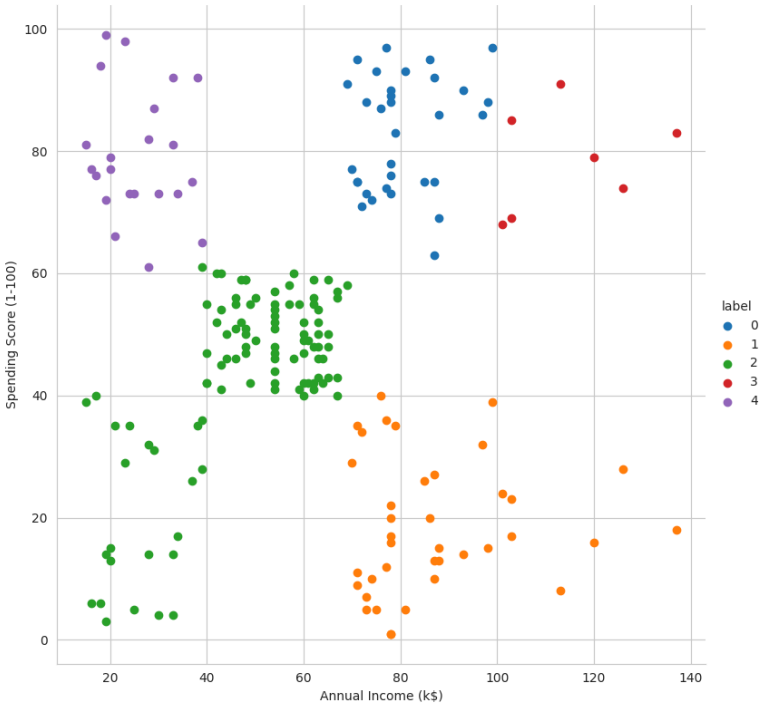
plt.show()
features_el = df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss = []
for i in range(1,10):
   model = KMeans(n_clusters=i)
   model.fit(features_el)
   wcss.append(model.inertia_)

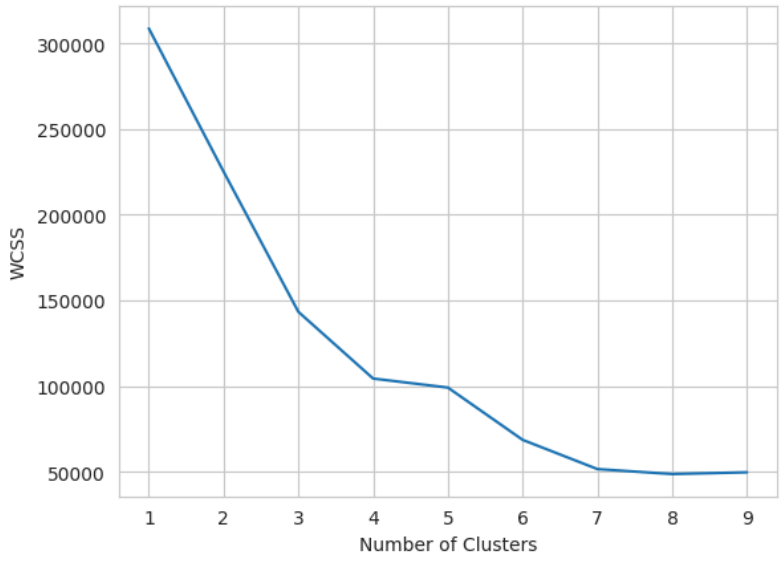
plt.plot(range(1,10),wcss)
plt.show()
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```







RESULT:

Thus the code is executed successfully.